

Kashmith Samaranayaka

Senior Software Engineer · AI/ML

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Professional Summary

Software Engineer with 2+ years building production **AI/ML and backend systems**, specializing in LLM-powered and agentic applications, MLOps pipelines, and scalable infrastructure. **Open-source core contributor to the Jaseci stack**, a major developer of the **jac-scale** Kubernetes deployment plugin and contributor to the **jaclang** language. Co-author of **two IEEE-published papers** on deep learning. Graduate of the **University of Moratuwa**.

Technical Skills

Languages: Python, Go, Jac (jaclang), SQL, Bash

AI/ML & GenAI: LLMs, RAG, Agentic AI (LangGraph), Fine-tuning (Unsloth), NLP, Deep Learning

ML Frameworks: PyTorch, TensorFlow/Keras, scikit-learn, Hugging Face

MLOps & Infra: Docker, Kubernetes, MLflow, Apache Airflow, CI/CD

Backend & Data: FastAPI, Jaseci, REST APIs, PySpark, Kafka, Redis Streams, Vector Databases (Chroma)

Experience

AI/ML Engineer

April 2024 - Present

Jaseci Lab, Ann Arbor, MI, USA

Remote

- Core contributor to the open-source **Jaseci stack**, a major developer of the **jac-scale** Kubernetes deployment plugin and contributor to the **jaclang** language.
- Developed and maintained backend infrastructure for **Tobu**, an AI-powered personal memory platform with integrated LLMs serving **1,000+ users**; took the system to production fast enough to support the company's **\$1.5M raise**.
- Built scalable PySpark data pipelines for user analysis and feature optimization, and shipped ML features for image analysis and conversation processing.

Data Scientist

April 2024 - June 2024

Utech Technologies (Pvt) Limited, Wattala, Sri Lanka

Remote

- Designed and deployed an end-to-end energy forecasting pipeline with MLflow experiment tracking and model versioning for reproducible, production-ready forecasts.
- Automated ML workflows with Apache Airflow for data ingestion, model retraining, and deployment.
- Built production API services with FastAPI and Docker for real-time energy prediction and dashboard visualization.

Application Engineer Intern

January 2023 - July 2023

Synopsys Lanka (Pvt) Ltd, Colombo, Sri Lanka

On-site

- Developed an NLP-based search engine with LSTM models for automated email classification.
- Automated large-file compilation and notification systems using Python, Bash, and CSH scripting.
- Managed 17 processor designs through custom automation scripts for daily performance assessment and resource optimization.

Education

B.Sc. Engineering (Hons.) in Electrical Engineering

Jan. 2020 - July 2024

University of Moratuwa

Moratuwa, Sri Lanka

- Overall GPA: 3.31 (Second Upper Class)** · Washington Accord-accredited program
- Dean's List – 2nd semester

Open-Source Contributions

Jaseci Stack: Core Contributor (jac-scale, jaclang)

- Core developer of **jac-scale**, the deployment plugin that scales identical local Jac code to Kubernetes production, auto-provisioning databases, Redis caching, and authentication with zero configuration.
- Contributor to **jaclang**, an AI-native language featuring graph-based Object-Spatial Programming with native Python, JavaScript, and C interoperability.

Projects

Industry & AI Projects

JacHammer - AI Agentic Software-Generation Platform

- Backend developer on an agentic AI platform that generates and deploys full-stack jaclang applications end-to-end, orchestrating LLM-driven code generation, build, and one-click deployment.

Tobu Lite - AI Image-to-Video Generation

- Built the image-to-video feature for the Tobu platform, a generative media pipeline integrating Google's **Veo 3** model that lets users create videos from their photos using selectable templates.

Deep Research Assistant - Agentic AI Application

- Built a production-grade agentic app for analytical question answering using LangGraph for multi-step reasoning, with web search and Chroma vector retrieval, deployed via Docker with a Streamlit interface.

Einstein Pro - AI Patent Generation & Document Management Platform

- Backend developer for an AI platform that generates patents and manages documents, including a RAG-based chatbot over a vector database for fast, accurate retrieval.

FriendZone - AI-Powered Memory Sharing Platform

- Full-stack AI platform to capture, organize, and share memories via conversational interfaces, with automatic image analysis (who/what/where/when) using the OpenAI API; backend built on jaclang and jac-cloud.

Self-Adaptive NILM using Deep Learning (Research)

- Designed WaveNet/CNN models with ensemble methods for appliance-level power prediction; applied pseudo-labelling and transfer learning for self-adaptation to appliance ageing without retraining.
- Co-authored and published the work at the **2024 IEEE World AI IoT Congress** (see Publications).

Publications

M. S. K. Nisakya, S. M. L. Arampola, W. A. Yasodya, S. Kumarawadu, V. Logeeshan, C. Wanigasekara. "Self-Adaptive Non-Intrusive Load Monitoring Using Deep Learning." *2024 IEEE World AI IoT Congress*, 2024. 

W. A. Yasodya, S. M. L. Arampola, M. S. K. Nisakya, V. Logeeshan, S. Kumarawadu, C. Wanigasekara. "Self-Adaptive Deep Learning Framework for Non-Intrusive Load Monitoring: Addressing Aging Appliance Challenges With Transfer Learning and Pseudo Labeling." *IEEE Access*, vol. 13, pp. 106524–106539, 2025. 

Professional Certifications

Machine Learning Specialization

DeepLearning.AI & Stanford University

TensorFlow: Advanced Techniques

Specialization

DeepLearning.AI

Certified Kubernetes Administrator (CKA) (In Progress)

Udemy

Machine Learning Engineering for Production (MLOps)

Specialization

DeepLearning.AI

Data Structures and Algorithms Specialization

University of California San Diego

AWS Solutions Architect – Professional (In Progress)

Udemy